

exactly the preferred “retirement” income pattern for a country with a SWF whose oil wells have run dry or a foundation whose benefactor has died. Surveys also indicate that most people prefer steady or rising real incomes, even in retirement: the declining consumption shares that we observe in data (see Figure 5.6) could be because people are forced to cut back consumption in retirement by a lack of money coming in and poor planning, not because they like replacement rates less than one.³⁹ Life-cycle models allow us to compute optimal pre-retirement savings and investment patterns for such investors by using special utility functions.

With *habit formation* utility, investors intensely feel the pain of being unable to live on in the style to which they’ve become accustomed. Countries and universities, for example, find it tough to cut spending below certain levels, and these levels (or habits) rise over time. Using habit formation in life-cycle models generates very smooth consumption patterns in late life.⁴⁰ The investor has to save a very large amount during her working life to finance this and continues to save after retirement (there is large wealth accumulation at all ages across the life cycle).

Ratchet Consumption

Dybvig (1995, 1999) develops a model of *ratcheting spending*, which should be attractive to investors wishing to maintain spending; it even allows spending to rise when investments have sufficiently appreciated.⁴¹ To sustain their spending pattern, investors would set aside some of their portfolios in a risk-free asset. This risk-free portion can be thought of as an insurance policy guaranteeing future minimum spending, which is called the *liability-hedging portfolio*. The remainder of the funds is invested in a (levered) stock portfolio. When the risky portfolio has sufficiently appreciated, the investor ratchets up spending and can protect it by transferring some of the appreciated monies into the risk-free portfolio.

While Dybvig’s framework is ideal for endowments, individuals, and even countries that do not wish to experience declines in their standards of living, current circumstances present two challenges in following ratcheting spending rules. First, interest rates now are very low (real rates are negative at the time of writing, see chapter 11), and the necessary minimum spending rates from Dybvig’s model are very, very low. Second, a large liability hedging portfolio is held in bonds—and expected returns on bonds are very small at the moment and potentially negative. Nothing, unfortunately, comes for free.

³⁹ See Beshears et al. (2012).

⁴⁰ Fuhrer (2000) and Michaelides (2002) were the first to put habit utility in life-cycle models. See also Polkovnichenko (2007).

⁴¹ Dybvig’s (1995, 1999) ratchet consumption model is a dynamic generalization of constant proportion portfolio insurance. This was invented by UC Berkeley finance professors Hayne Leland and Mark Rubinstein, which they marketed through their firm LOR (Leland–O’Brien–Rubinstein). It was blamed for a large part of the 1987 stock market crash. See Bernstein (1992) for a history. Riedel (2009) generalizes Dybvig’s original formulation, allowing for jumps and fat tailed processes.